HOUSEHOLD WATER USAGE AND BILLING MONITOR

Monitor • Manage • Save Water • Save Money

Every Drop Counts

ABSTRACT

The Household Water Usage and Billing Monitor is a smart system that tracks water consumption in real-time and calculates the bill based on the usage and applicable rates. It helps households monitor their water usage, avoid wastage and ensures accurate and transparent billing.



KEY FEATURES

- **Real-time Water Usage Tracking**
  Monitors consumption using a water meter (manual/automatic).
- **Automatic Bill Calculation**
  Calculates bill based on usage slabs and predefined rates.
- **Usage History & Reports**
  View daily, weekly and monthly consumption with graphs.
- **Leakage & High Usage Alerts**
  Notifies users about abnormal usage or possible leaks.
- **User Friendly Interface**
  Easy to use and accessible on mobile and web platforms.

HOW IT WORKS

1. **Read Meter Data** — Collects current water usage.
2. **Process Usage Data** — Stores and calculates consumption.
3. **Calculate Bill** — Applies tariff rates and extra charges.
4. **Generate Bill** — Creates a detailed bill statement.
5. **Notify User** — Shows usage, bill and alerts via app/portal.

SYSTEM OVERVIEW



BILLING EXAMPLE

Usage Slab (Litres)	Rate (₹ per 1000 L)	Amount (₹)
0 – 5000	5.00	25.00
5001 – 10000	8.00	40.00
10001 – 15000	12.00	60.00
> 15000	15.00	75.00

Total Water Bill = ₹ 200.00

BENEFITS

- Helps save water and reduce wastage.
- Ensures accurate and transparent billing.
- Promotes responsible water usage.
- Improves convenience and efficiency for households and water providers.

IMPACT

- Conserves Water Resources
- Reduces Household Expenses
- Builds Awareness
- Supports Sustainable Future

TECHNOLOGY USED

- IoT Sensors (Flow Meter)
- Python / Java (Backend)
- MySQL (Database)
- Web & Mobile App

Smart Monitoring • Fair Billing • Sustainable Living